EVK Quick Start Guide
Power Consumption Evaluation

Revision History

Date	Version	Description
July 16, 2019	0.40	Initial version created.
November 20, 2020	0.41	Updated various sections.
December 21, 2020	0.42	Added footnote under Overview.
March 30, 2021	0.43	Updated Figure 9, Table 2.
April 14, 2021	0.44	Updated format, no content change.
July 1, 2021	0.45	Added Figure 3 for ATM2251.

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Overview

The ATM22x1 Evaluation Board (EVB) is pre-configured to transition through various active and low power states[1]. This guide provides instructions for Evaluation Kit (EVK) users to perform a power consumption evaluation of the ATM22x1.

EVK Setup

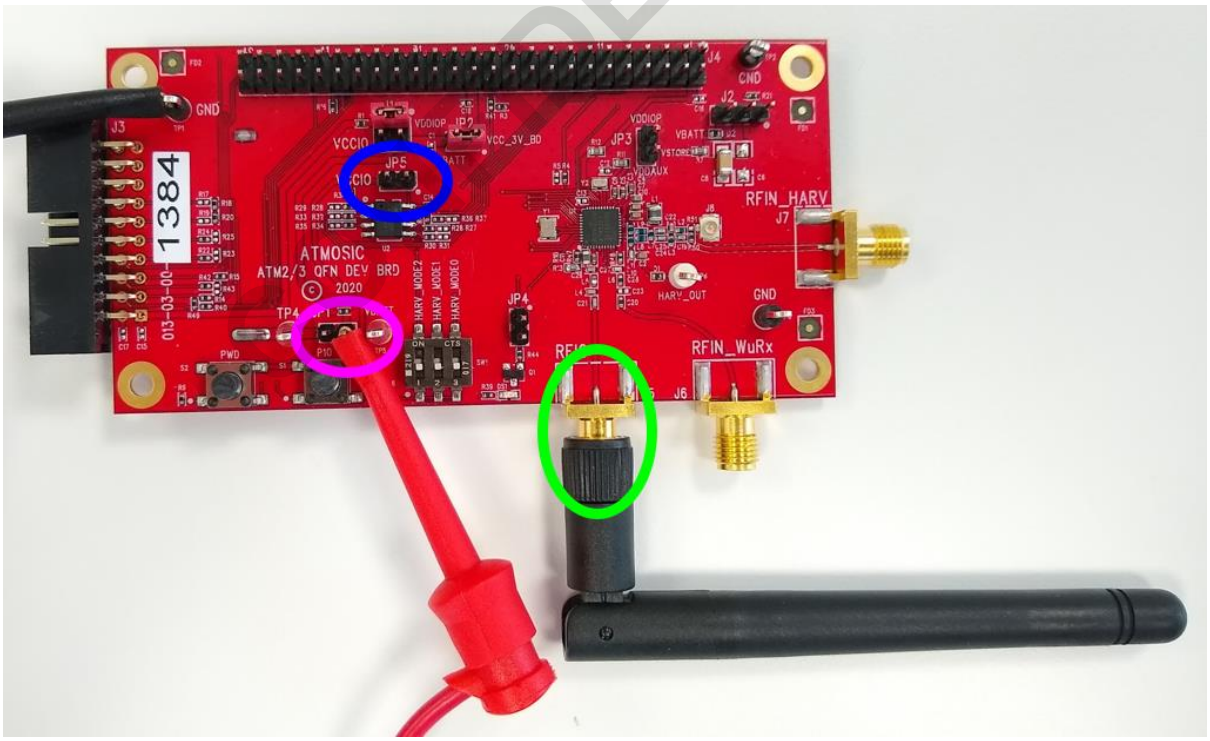
Figure 1 shows the power consumption setup for revisions C2a and D2a of the ATM2201 EVB.

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[1] To configure ATM32xx and ATM2202 EVB's in a similar fashion, please contact Atmosic Support.



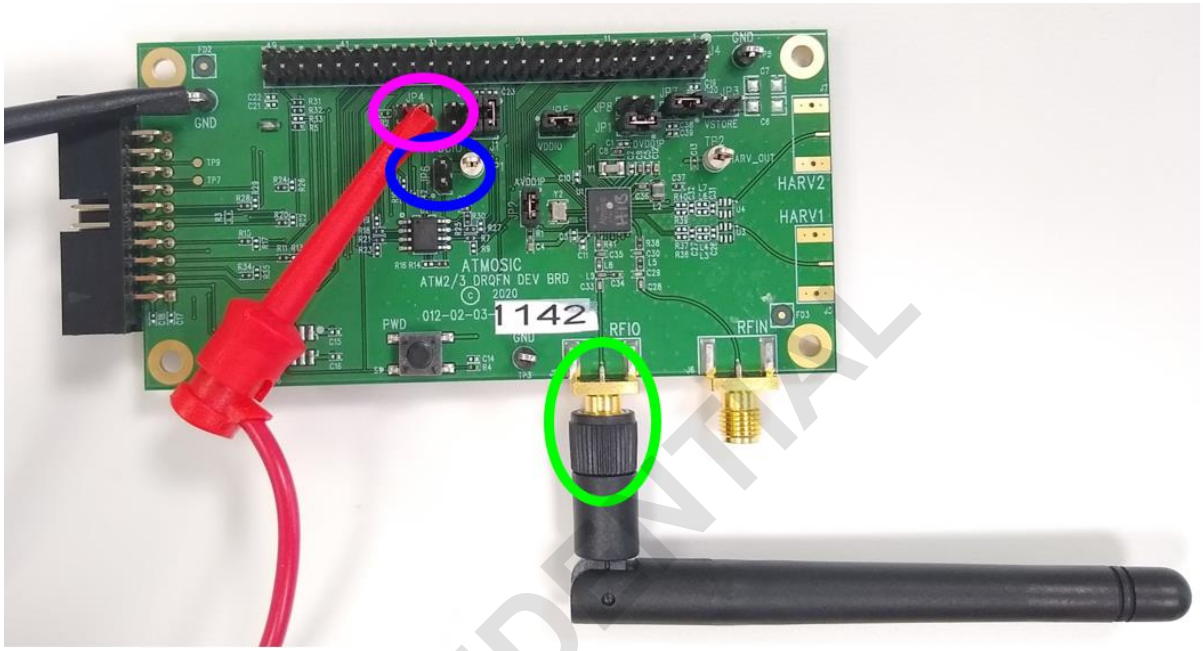
ATM2201 EVB Revision



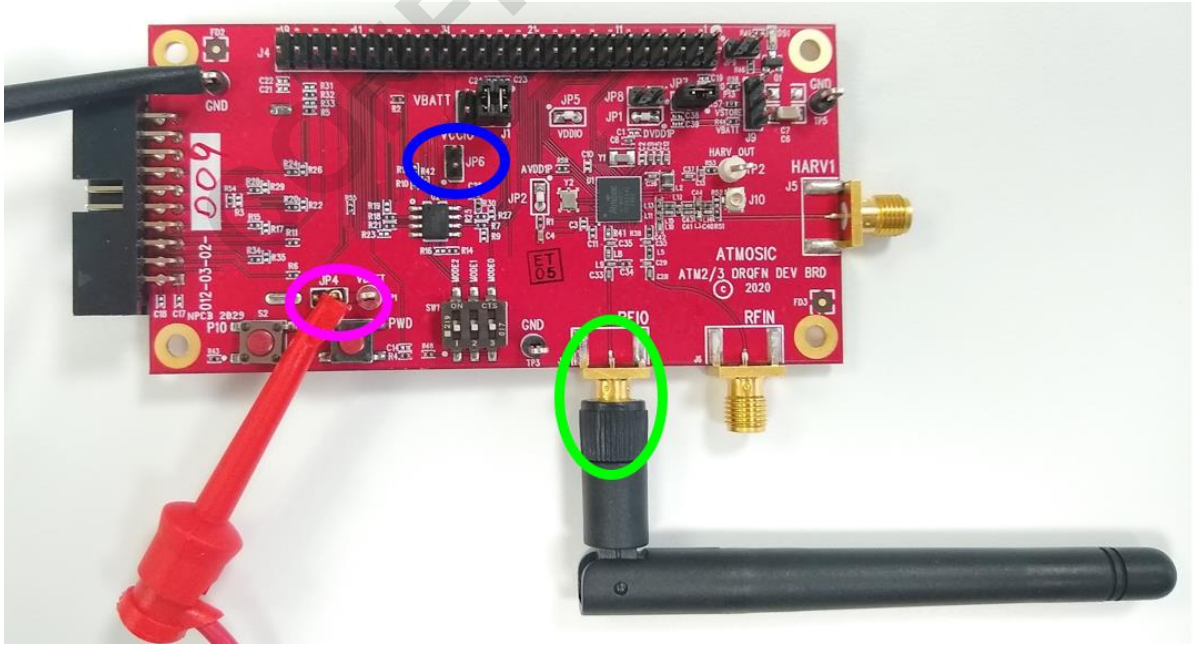
ATM2201 EVB Revision D2a

Figure 1 - ATM2201 EVB Revision C2a/D2a

[Figure 2](#) shows the power consumption setup for revisions C2a and D3a of the ATM2221 EVB.



ATM2221 EVB



ATM2221 EVB

Figure 2 - ATM2221 EVB Revision C2a/D3a

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Figure 3 shows the power consumption setup for revision A1 of the ATM2251 EVB.

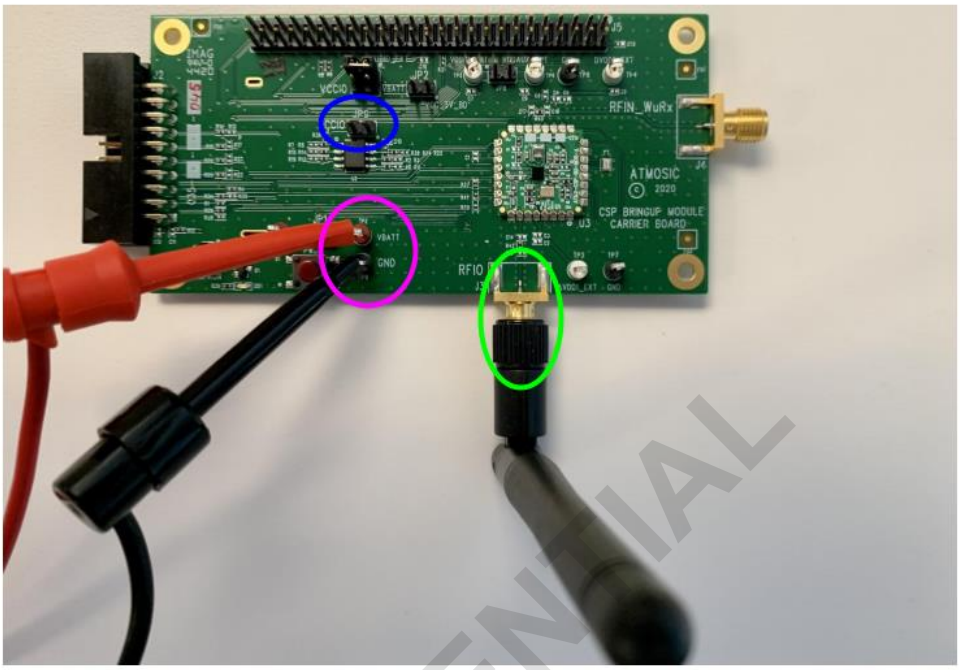


Figure 3 - ATM2251 EVB Revision A1

The same instructions apply to all of the above revisions of EVB:

1. Attach the 2.4 GHz antenna to the RFIO port of the EVB (circled in **green**).
2. Open the external flash supply jumper (circled in **blue**).
3. Connect a 3 V power supply to pin 1 of the coin cell battery jumper (circled in **magenta**) and any ground (GND) test clip.
4. A DC power analyzer, 6-1/2 digit multimeter, or 10 Ω resistor and oscilloscope can be used for power measurements.

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EVB Configuration

The EVB is pre-configured to run a beacon application from internal ROM with the OTP configuration parameters shown in Table 1. It will advertise at a 1s interval for 30s (advertising phase) and then hibernate for the next 30s (hibernation phase) as shown in Figure 4. During the advertising phase, the ATM22x1 goes into retention between the 1s beacons.

Table 1 - Beacon Configuration in OTP

Output Power	0 dBm
# of advertising Channels	3
Scannable	Yes
Connectable	No
Type	Eddystone (39 byte PDU)
Local Name	Shortened (in advertisement): Axxxxxx Complete (in scan response): ATM22x1 E3d EVB xxxxxx xxxxxx: last 6 digits of board address

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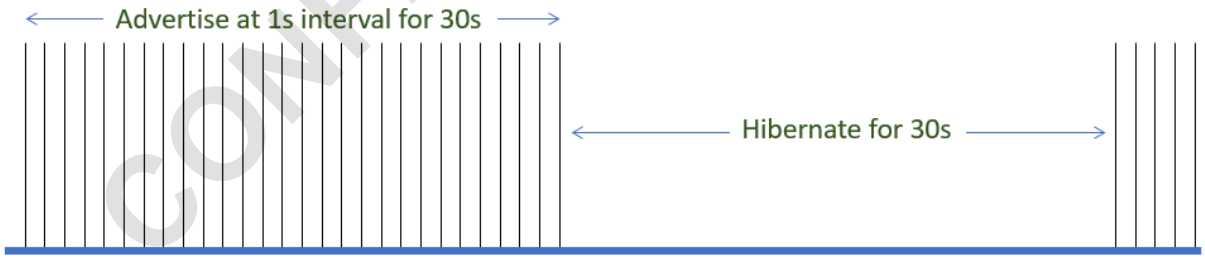


Figure 4 - Typical Beacon Cycle's Power Profile At VBATT

Power Measurement Procedures

Average Power Measurement with Multimeter

1. Connect a 6-½ digit multimeter in series with the 3 V supply.
2. Set the multimeter for DC Current and increase averaging to as long as possible. For example, the Keysight 34465A averaging time can be set to 1s.
3. The current should typically be around 6.8 µA during the advertising phase and drop to about 0.7 µA during the hibernation phase. Note that a multimeter will not show the power profile. See Figure 5.



Figure 5 - Average Power Measurement with Multimeter

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Approximate Power Profile Measurement

1. Connect a 10 Ω resistor in series with the ground of the EVB.
2. Put an oscilloscope probe across the 10 Ω resistor.



Figure 6 - Approximate Power Profile Measurement

3. The oscilloscope will show a periodic beacon profile (3 pulses for 3 channels) during the advertising phase followed by the hibernation phase without beacons.
4. The approximate maximum transmit power can be approximated by dividing the measured peak voltage across the 10 Ω resistor. It should be approximately 2.5 mA. Note that the oscilloscope will not provide enough resolution to measure other power states.

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More Accurate Power Profile Measurement

Measuring the dynamic current more accurately requires a DC power analyzer such as the Keysight N6705C. For the Keysight N6705C, it is important to use auto-ranging and the maximum number of horizontal data points to observe the most accurate power profile. Figures 7 to 11 show current consumption measurements of various states.



Figure 7 - Transmit Current Profile

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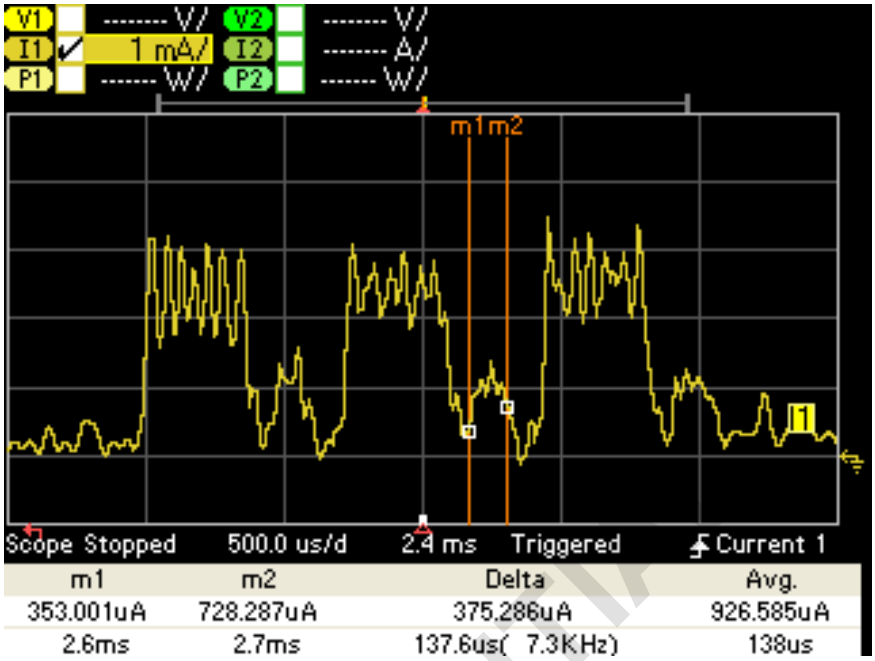


Figure 8 - Receive Current Profile

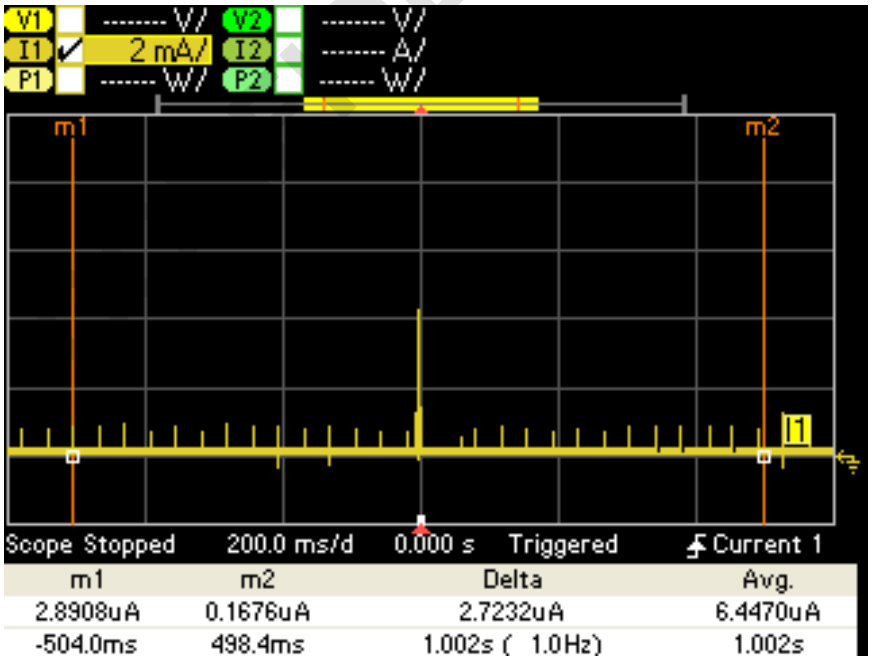


Figure 9 - Current Profile Over 1s Interval During Advertising Phase

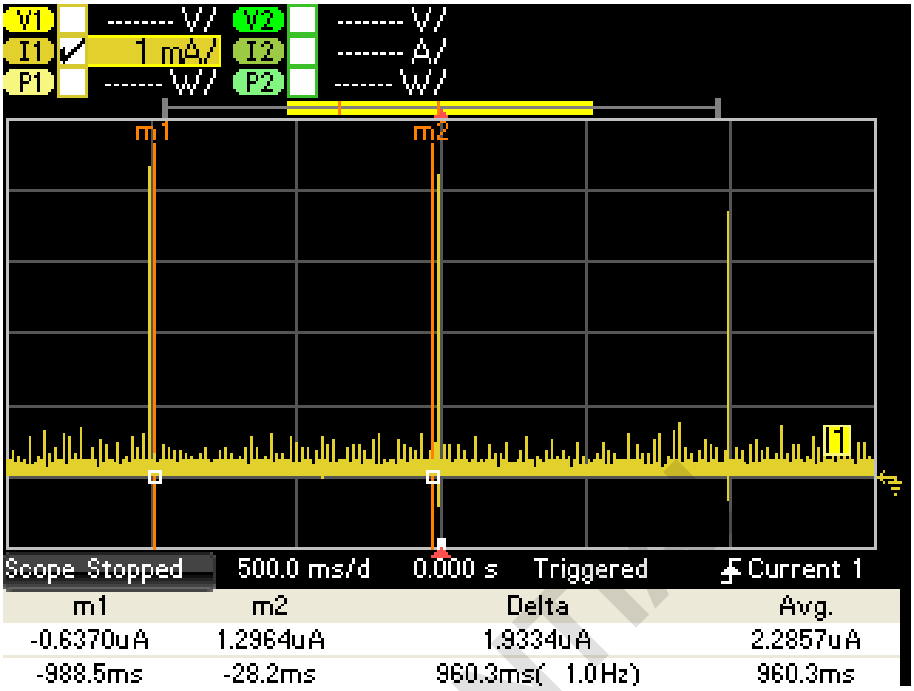


Figure 10 - Retention Current Profile

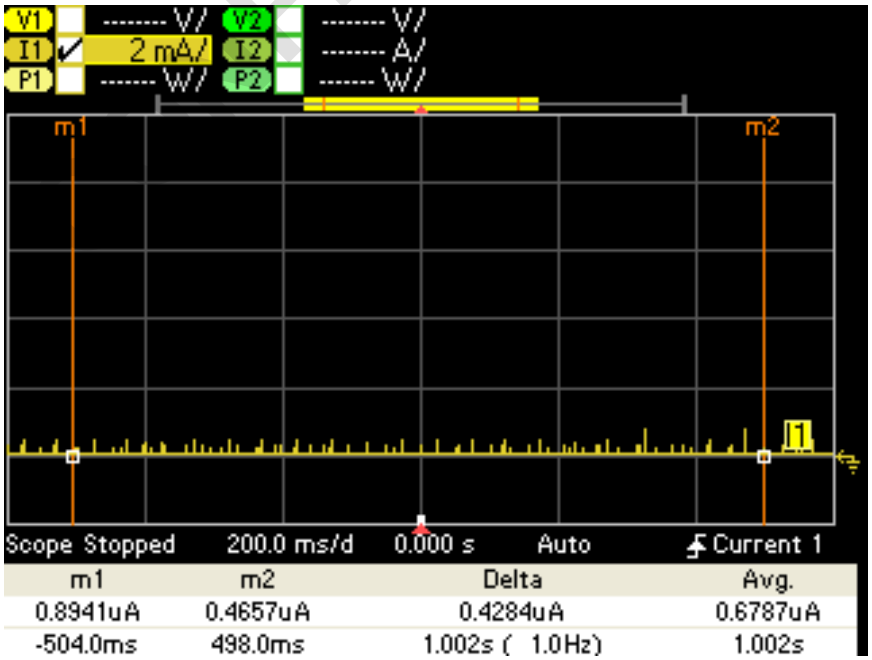


Figure 11 - Hibernation Current Profile

Table 2 summarises typical EVB current measurements for each of the above states.

Table 2 - Typical Current for Various States

State	Average current on VBAT = 3 V (µA)
Transmit (0 dBm)	2500
Receive	1000
1s interval during advertising phase	6.8
Retention	2.3[2]
Hibernation	0.7

It is normal to observe small peaks during low power or active modes such as the ones in Figure 10 and Figure 11. These peaks result from typical operation of the DC/DC switching regulator and are generally harmless in most applications.

Further Power Reduction in Low Power States

The default power management settings for the ATM2/ATM3 are optimized to reduce supply transient current spikes. However, the hibernation and retention power consumption can be further reduced if the DVDD1P supply is lowered dynamically. A side effect of lowering the voltage is a current spike due to charging of the supply bypass capacitor on DVDD1P whenever the ATM2/ATM3 exits hibernation or retention. The energy from the current spike is not lost and is used to power the ATM2/ATM3 when it re-enters hibernation or retention.

Figure 12 shows an advertisement interval with DVDD1P lowered to 0.8 V during retention. When compared to Figure 9, the average current consumption is lower because of reduced retention current. The max current is higher because of the current spike. Please note that this measurement was made with a 4.7 uF capacitor on DVDD1P.

Please contact your support team if you are interested in exploring further power reduction in low power states.

[2] Retention current is 2.6 µA for ATM22x1-x0x. Refer to Atmosic Chip Revisions Application Note for information regarding chip revisions.

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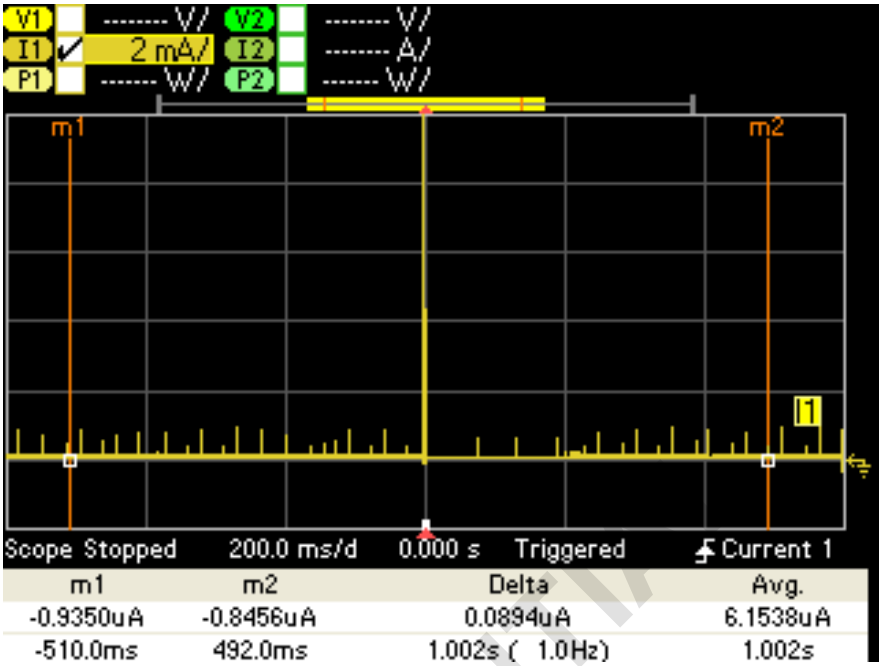


Figure 12 - Current Profile Over 1s Interval During Advertising Phase With Lower DVDD1P

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